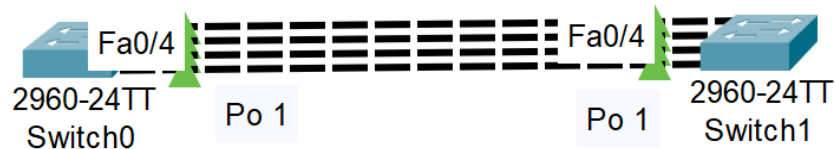


# Etherchannel:

EtherChannel is a link aggregation technology developed by Cisco that combines multiple physical Ethernet links into a single logical link. This technology provides several benefits, such as increased bandwidth, redundancy, and load balancing. Here's a detailed explanation of how EtherChannel works and its key features:

## Key Features of EtherChannel:

1. Increased Bandwidth (Combine Multiple Links to behave like single link)
2. Redundancy
3. Scalability
4. Upto 8 Links can be Aggregate
5. Combined to balance the traffic (Loadbalancing)



# In Etherchannel we aggregate the link:

A Port-channel will be formed which will work as a single link by combining all links and speed can be double/Triple/Quadrupled etc.

**Port-channel 1**

*(Fa0/1+Fa0/2)*



***BW=100+100=200 MBPS***

**Port-channel 1**

*(Fa0/1+Fa0/2)*

# Types of Etherchannel:

## Static Etherchannel:

Configured manually by the network administrator. This method does not use any negotiation protocols and requires consistent configuration on both ends of the Etherchannel link.

```
Switch(config)#interface range fa0/1-4
```

```
Switch(config-if-range)# channel-group 1 mode on
```

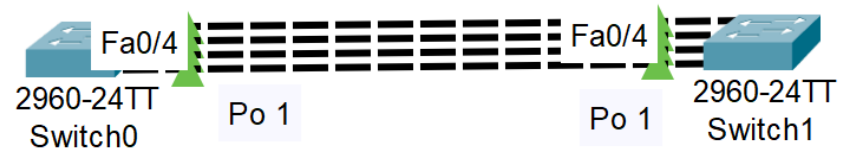
```
Switch(config-if-range)# exit
```

```
Switch(config)# interface port-channel 1
```

```
Switch(config-if)# Switchport mode trunk
```

```
Switch(config-if)# exit
```

```
Switch(config)# end
```



# Dynamic EtherChannel:

Uses negotiation protocols to automatically form an EtherChannel.  
The two main protocols are:

**Port Aggregation Protocol (PAgP):** A Cisco proprietary protocol that automatically negotiates the creation of an EtherChannel link.

**Link Aggregation Control Protocol (LACP):** An IEEE 802.3ad standard protocol that performs a similar function to PAgP but is vendor-neutral, allowing interoperability between different vendors' equipment.

# Dynamic Etherchannel:

<https://www.youtube.com/@techwithravish>

Dynamic EtherChannel is a method of creating an EtherChannel link that uses a negotiation protocol to automatically manage the bundling of multiple physical links into a single logical link.

The two main protocols used for dynamic EtherChannel are:

- 1. Port Aggregation Protocol (PAgP)**
- 2. Link Aggregation Control Protocol (LACP)**



# Possible Combination of Etherchannel:

| Mode (SW 1)             | Mode (SW 2)             | Result:                      |
|-------------------------|-------------------------|------------------------------|
| ON                      | ON                      | Etherchannel(Static)         |
| <i>Active (LACP)</i>    | <i>Active (LACP)</i>    | <i>Etherchannel(Dynamic)</i> |
| <i>Active (LACP)</i>    | <i>Passive(LACP)</i>    | <i>Etherchannel(Dynamic)</i> |
| Passive (LACP)          | Passive(LACP)           | No Etherchannel(Dynamic)     |
| <i>Desirable (PAgP)</i> | <i>Desirable (PAgP)</i> | <i>Etherchannel(Dynamic)</i> |
| <i>Desirable (PAgP)</i> | <i>Auto(PAgP)</i>       | <i>Etherchannel(Dynamic)</i> |
| Auto (PAgP)             | Auto (PAgP)             | No Etherchannel(Dynamic)     |

# Link Aggregation Control Protocol (LACP)

LACP is an open standard (IEEE 802.3ad) protocol that provides similar functionality to PAgP but is vendor-neutral, allowing it to work across different manufacturers' devices.

LACP can operate in two modes:

**Active:** Actively attempts to negotiate with the other side to form an EtherChannel.

**Passive:** Passively waits for the other side to initiate the negotiation.



# *LACP Configuration in Etherchannel:*

```
Switch(config)# interface range fa0/1-2
```

```
Switch(config-if-range)# channel-group 1 mode active
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface port-channel 1
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# exit
```

```
Switch(config)# end
```

Port-channel 1 (Fa0/1+Fa0/2)



Port-channel 1 (Fa0/1+Fa0/2)



# Port Aggregation Protocol (PAgP):

PAgP is a Cisco proprietary protocol used to automatically negotiate and form an EtherChannel link between switches.

PAgP can operate in two modes:

**Desirable:** Actively attempts to negotiate with the other side to form an EtherChannel.

**Auto:** Passively waits for the other side to initiate the negotiation.



# *PAgP Configuration in Etherchannel:*

```
Switch(config)# interface range fa0/1-2
```

```
Switch(config-if-range)# channel-group 1 mode desirable
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface port-channel 1
```

```
Switch(config-if)# Switchport mode trunk
```

```
Switch(config-if)# exit
```

```
Switch(config)# end
```

Port-channel 1 (Fa0/1+Fa0/2)



Port-channel 1 (Fa0/1+Fa0/2)